

HANISHA VEMIREDDY

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PROFESSIONAL SUMMARY

Data Scientist with 8+ years of experience across academia and industry, specializing in experiment design, predictive modeling, and delivering analytics solutions with measurable business impact. Proven ability to translate complex data into clear, actionable insights for stakeholders across agriculture, biotech, financial services, and consulting. Experienced in building scalable pipelines and applying ML and statistical methods to drive cost savings, optimize processes, and support strategic decision-making.

SKILLS

Programming: Python, R, SQL, MATLAB, JMP

ML & Statistics: Experimental Design, Causal Inference, Linear Mixed Models, Hypothesis Testing,, Scikit-learn, XGBoost, Neural Networks, Random Forests, Time Series Analysis, Outlier Detection (Isolation Forest, LOF)

AI & Modeling: Transformer Models, Time Series (TimesFM), Feature Engineering, Model Evaluation

Data & Visualization: Pandas, Tidyverse, R Shiny, GGPlot, Spotfire, Power BI, Tableau

Cloud & Tools: AWS, Snowflake, Git, GitHub Copilot, Claude Code

WORK EXPERIENCE

Data Scientist II | **Corteva Agriscience** | Remote 2022 - 2025

- Drove \$1M+ in cost savings by designing experiments and hypothesis tests that optimized manufacturing productivity across multiple scales.
- Built reusable ETL pipelines and analytics frameworks, establishing key metrics to identify scalability bottlenecks and optimize data management.
- Leveraged linear mixed models to develop predictive analytics informing strategic decisions and identifying high-value business opportunities.
- Built multivariate outlier detection models for sensor data that identified bioprocess deviations early, enabling timely interventions.

Graduate Research Assistant | **Iowa State University** | Ames, IA 2017 - 2022

- Improved soybean yield prediction accuracy by 40% through innovative training data imputation methodologies.
- Optimized test location selection, improving advancement decisions by 30% using multi-environmental historical data.
- Achieved 95% empirical coverage for genotype ranking using a bootstrap-based approach, robust to up to 95% missing data.

Data & Operations Research Intern | **Principal Financial Group** | Des Moines, IA 2019

- Automated quality validation pipeline for financial sentiment data using Python and SQL (Snowflake) to enhance investment decision-making.
- Created Power BI dashboards to present complex financial data in clear, stakeholder-ready visualizations.

Data Science Consultant | **Widasconcepts Pvt Ltd** | Bengaluru, India 2016 - 2017

- Designed predictive models leveraging GPS behavioral data to analyze customer preferences and drive personalization strategies.

EDUCATION

PhD, Industrial Engineering | **Minor in Applied Statistics** | Iowa State University, Ames, IA 2017 - 2022

Masters, Quality Management Sciences | Indian Statistical Institute, Bengaluru, India 2014 - 2016

Bachelors, Mathematics | Indian Statistical Institute, Bengaluru, India 2011 - 2014